

Chapter 3

Ethics and Ethical Analysis¹

Introduction of Ethics²

Data Ethics (Datacamp)³

By Dr. Ali Algarni

¹ Ethical and Social Issues in the Information Age

² Ethics for the information age

³ <https://www.datacamp.com/blog/introduction-to-data-ethics>

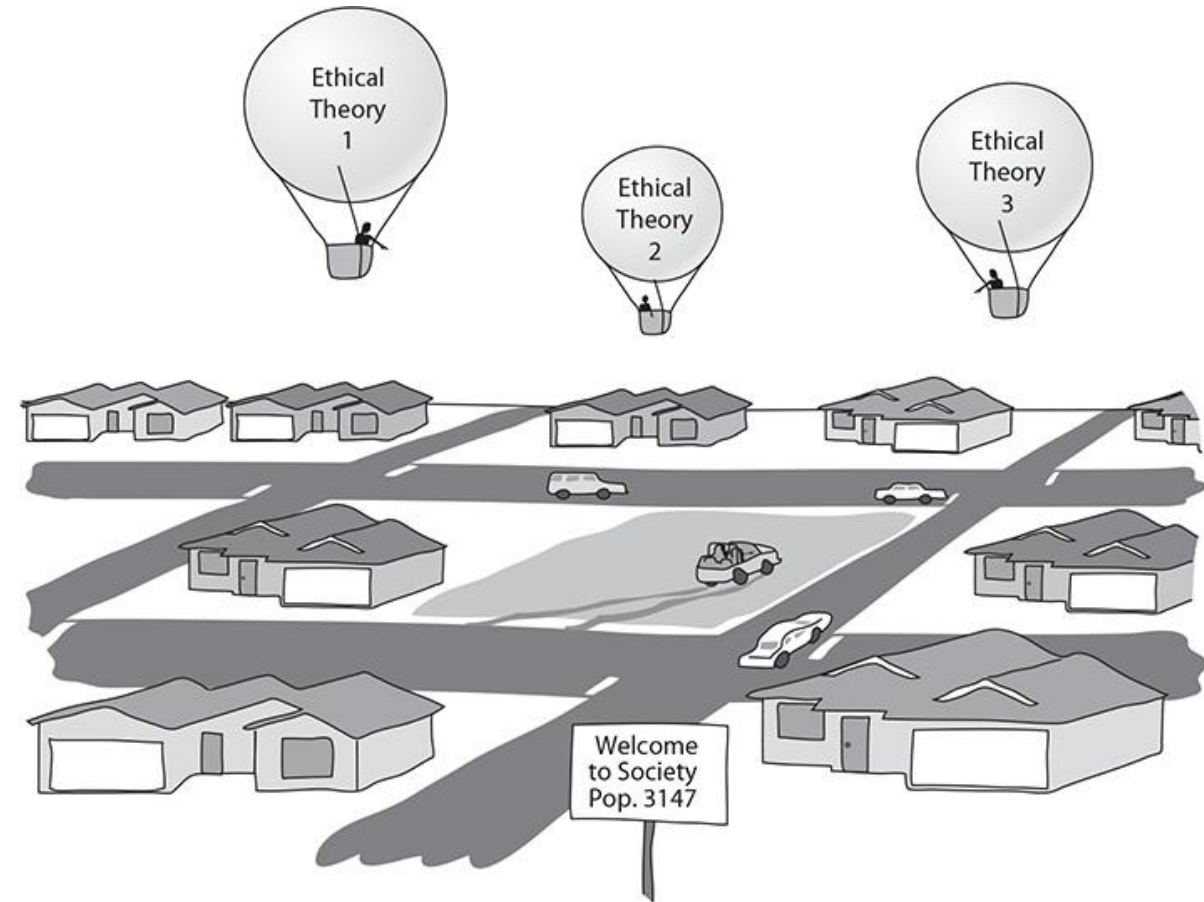
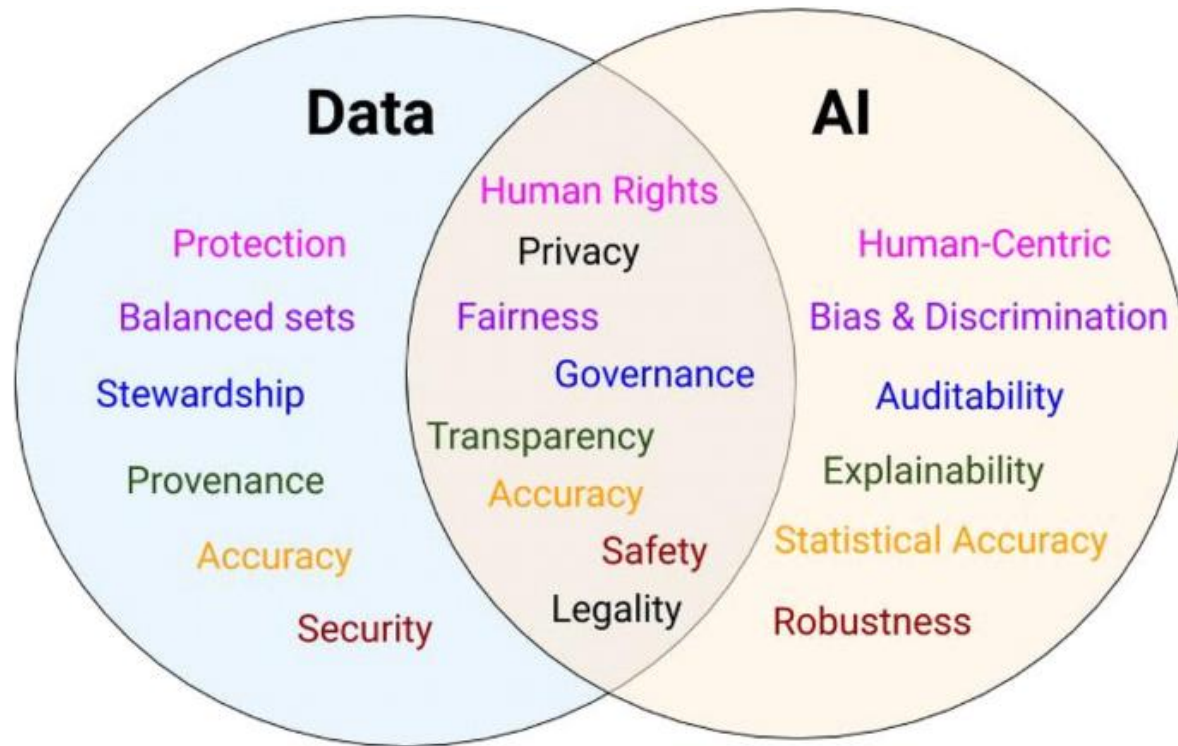
Outlines

- Law
- Ethical Theories (6 theories)
- Codes of Ethics
- Ethical Decision Making and Reasoning
- Ethical and Legal Framework for Information
- Anonymity, Security, and Privacy
- Project

Recap

- Society
 - Universal, local, online communities (Zoom, Minecraft, Reddit, Discord)
- Morality
 - guidelines, tools, equipment, traffic light, email standard, integrity statement
- Ethics
 - Based on morality
 - Right or wrong? Good or bad? Guilty or not?
- Law?
 - Regulations

Recap



Law

- Rules of conduct or
- Actions recognized by customs or decreed by a formal body and enforceable by some instrument.
- Laws are derived from moral codes
- We obey two types of laws: Natural and Conventional

Nature Law (Moral Law)

- Unwritten but universal
- Consists of rights:
 - Self-defense (preservation)
 - Individual property
 - Liberty
- Before organized human societies, humans existed because of natural law.
- Civilization is based on it.
- **Human dignity (data-driven technologies)**

Conventional Law (Physical Law)

- It is a system of rules created by and for human beings – through, though not always, public deliberations
- It varies from society to society
- Its purpose is to:
 - Protect human life, property and liberty.
 - prescribe a system or punishments for unlawful acts – Penal Code

Ethical Theories

- Human actions have been judged good or bad/right or wrong
 - based on theories/systems/justice developed by society
- Codes of ethics have then been drawn up based on these ethical theories.
- The processes of reasoning, explanation, and justification based on these theories/systems

Ethical Theories

- Consequentialism
- Deontology
- Human Nature
- Relativism
- Hedonism
- Emotivism

1- عواقب
2-
3-

4-
5-
6-

Consequentialism (Ethical theory)

- Actions are judged good/bad depending on the outcome/ results (consequences) of such actions
- There are three types of Consequentialism:
 - Egoism- putting an individual's interests and happiness above everyone else's.
 - Utilitarianism – putting a group's interests and happiness above self.
 - Altruism – actions are judged good if the consequences are favorable to all except the actor.

Deontology-duty theory. (Ethical theory)

- an **action** is good if done as a duty.
- an act is considered good if the individual committing it had a good reason to do so
- Example,
 - killing vs killing to save lives
 - Hacking vs Ethical hacking
 - data breach vs Finding areas where sensitive data could be compromised in a cyber attack

Human Nature. (Ethical theory)

- Actions are judged good based on the **capabilities** of the actor (guilty by reasons of insanity)
- We are supposed to discover and then develop those capabilities and use them as benchmarking
- Example,
 - an individual committing an evil action is lacking in some capabilities
 - Decision making involves people with no technical or analytical skills

Relativism. (Ethical theory)

- this takes right and wrong to be **relative to society** (hence no universal norms).
- Relativism also states that moral norms are not fixed in time.
- Example,
 - Marrying from family (royal)
 - Accessing children's artifacts and phones (data) in KSA vs US



Robbed
society

Hedonism (Ethical theory)

- Hedonism theory seeks maximum pleasure for all...
- It claims that pleasure is the only good thing in human life.
- Ethical hedonism claims that people ought to seek pleasure, and that pleasure is the moral good.

Emotivism (Ethical theory)

- Emotivism states that ethical statements are neither true nor false but depend on individuals.
- They are really statements about how someone feels
- Example, Restrictions on gun ownership are bad,
 - Because Restrictions on guns make me feel unsafe and worried
- Data collection via survey (bother people?)
 - Using machine to deal with patient (elders), bank transaction → make them nervous

Ethical theories (Summary)

- These theories are used as engines to help understand and justify human actions
- Theories have not changed with time and technology although the premises for human actions have changed
- Theories are used in layers of reasoning to justify all human actions.

Ethical theories (Summary)

Ethical Theories	Ethical Decision is Based on
1. Consequentialism	outcome/ results of such actions.
2. Deontology	Is the action done as a duty ?
3. Human nature	The capabilities of the actor
4. Relativism	be relative to society (hence no universal norms).
5. Hedonism	Seeks maximum pleasure for all.
6. Emotivism	Ethical statements are neither true nor false but depend on individuals .

Ethical theories (Summary)

- As data scientist, think about each theory when
 - Collecting
 - Analyzing
 - Storing
 - Disseminating
- For example,
 - When running ML/AI: The ethical theory of consequentialism can explain why the model's deployment is relevant to its moral evaluation.
 - *Difficulties in Comprehensive Risk Assessment in Real-World Environments*¹

1. A Conceptual Framework for Ethical Evaluation of Machine Learning Systems (<https://arxiv.org/html/2408.10239v1>)

Ethical Reasoning and Decision Making

- Reasoning and Logic → daily human interactions
 - *Cognitive processes* → based on experience, interpretation, foreseeing, ordering, analyzing, valuing, and making connections
 - *Logic* → distinguishing between truth and falseness.
- **Ethical decision making** is the process of making a decision which may result in one or **more moral conflicts**
 - Recognize the inherent ethical conflicts
 - Know the parties involved.
 - Understand how the decision will be implemented
 - The impact of the decision of the parties involved

Ethical Decision-Making Framework

- Such process involved in decision making **framework**, which includes:
 - Recognizing inherent ethical conflicts through comprehension, appreciation, and evaluation of all ethical dimensions of problem.
 - Understanding the problem and the facts of the problem.
 - Knowing the parties involved.
 - Being aware of alternatives.
 - Demonstrating knowledge of ethical practices.
 - Understanding how the decision will be implemented and who will be affected.
 - Understanding the impact the decision will have on the parties affected.
 - Understanding and comprehending the impact of the decision of the parties involved.

Codes of Ethics

- defined Ethics that are governed by a particular and definitive regiment of rules is called “**code of ethics**”.
- Although the use of codes of ethics is still limited to professions and high-visibility institutions and businesses, there is a growing movement toward widespread use.
- The wording, content, and target of many codes differ greatly.
- Some codes are written purposely for the public; others are targeting employees, and yet others are for professionals only.

Codes of Ethics

- Depending on the domain, ethical codes can take any of the following forms:
 - **Principles:** which may act as guidelines, references, or bases for some document
 - **Public policies:** which may include aspects of acceptable behavior, norms, and practices of a society or group.
 - **Codes of conduct:** which may include ethical principles.
 - **Legal instruments:** which enforce good conduct through courts.

Codes of Ethics: Example

- Association of Computing Machinery (ACM) Code of Ethics and Professional Conduct

On October 16, 1992, ACM's Executive Council voted to adopt a revised Code of Ethics. The following imperatives and explanatory guidelines were proposed to supplement the Code as contained in the new ACM Bylaw 17.

- **ACM Ethical and Social Issues in the Information Age (Pages 41-49)**
 - 24 imperatives
 - Example 3.3
 - “Organizational leaders are responsible for ensuring that computer systems enhance, not degrade, the quality of working life. When implementing a computer system, organizations must consider the personal and professional development, physical safety, and human dignity of all workers. Appropriate human–computer ergonomic standards should be considered in system design and in the workplace”.

Codes of Ethics Objectives

- Different domains and groups of people formulate different codes of ethics, but they all have among them the following objectives:
 - **Disciplinary:** By instilling discipline, the discipline, the group or profession ensures professionalism and integrity of its members
 - **Advisory:** a good source of tips to members and offer advice and guidance in
 - **Educational:** Ethical codes are good educational tools for members of the domain, especially the new ones
 - **Inspirational:** the codes should also carry subliminal messages to those using them to inspire them to be “good.”
 - **Publicity:** One way for professions to create a good clientele is to show that they have a strong code of ethics, and therefore, their members are committed to basic values and are responsible.

Codes of Ethics

Code of Ethics
vs
Code of Conduct

must
know
Search

Technology and Values

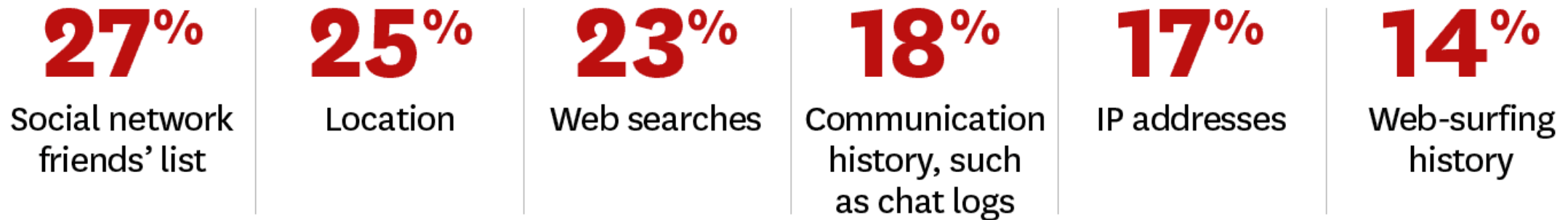
- New technologies are introduced to improve life
 - Some fail quickly; others become revolutionary
- New moral principles and ethical values are needed
- Successful technologies reshape society and raise moral and ethical concerns
 - Technological success often weakens individual moral values
- Global cultural, political, and social values are changing rapidly
 - The speed of change overwhelms society's ability to adapt
 - Society cannot develop and implement these values quickly enough

Data Ethics (*datacamp.com*)

In the Dark About Data

While most people are broadly aware that companies collect data on them, they're surprisingly uninformed about the specific types of data they give up when they go online.

Percentage of people who realize they're sharing their:



SOURCE TIMOTHY MOREY, THEODORE "THEO" FORBATH, AND ALLISON SCHOOP
FROM "CUSTOMER DATA: DESIGNING FOR TRANSPARENCY AND TRUST," MAY 2015

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Data Ethics (*datacamp.com*)

- **In September 2018**, hackers injected malicious code into British Airways' website, diverting traffic to a fraudulent replica site. Customers then unknowingly gave their information to fraudsters, including login details, payment card information, address, and travel booking information.
- **In March 2021**, the privacy of over 533 million Facebook users was compromised when their data was posted on an open hackers' forum. It was one of the largest data breaches of all time. The incident raised concerns about how organizations store and secure personal information and whether they should be allowed access to such data in the first place.

Big Data Ethics Principles (*datacamp.com*)

- **Transparency:** Clear communication about data collection, storage, and sharing practices is vital.
- **Accountability:** Organizations must take responsibility for the data they collect, including protecting it from breaches and misuse.
- **Individual Agency:** Individuals should have control over their personal data, including the ability to access, correct, or delete their information.
- **Data Privacy:** Privacy is the expectation that personal data will be protected from unauthorized exposure.

References:

- SDAIA AI Ethics Principles
- SDAIA Laws and Regulations (3rd and 4th part)
- General Data Protection Regulation(GDPR)
 - 2nd Chapter: <https://gdpr-info.eu/chapter-2/>

Chapter 5

Anonymity, Security, Privacy, and Civil Liberties¹

By Dr. Ali Algarni

¹ Ethical and Social Issues in the Information Age

Anonymity, Security, Privacy, and Civil Liberties

- Information has increased in value
- The demand for information is high due to:
 - High digitalization of information and increasing bandwidth.
 - Declining costs of digital communication
 - increased miniaturization/mobility of portable computers and other communications equipment.
 - Greater public awareness by the news media of the potential abuse of digital communication, especially the Internet.
- The danger for misuse is real

Anonymity

- What is “Anonymity” ?
 - Absence of identity
 - The state of being nameless, having no identity.
- It is extremely difficult for anybody to live a meaningful life while one is totally anonymous.
- There are two common types of anonymity:
 - Pseudo identity.
 - Untraceable identity.

Anonymity

- Pseudo Identity
 - An individual is identified by a certain pseudonym, code, or number (similar to a writer's pen name).
 - The most common variant of anonymity.
- Untraceable Identity
 - One is not known by any name including pseudo names.

Anonymity on the Internet

- The nature of the Internet, with its lack of political, cultural, religious, and judicial boundaries, has created opportunities for all faceless people to come out in the open.
- In particular, the Internet provides two channels through which anonymous acts can be carried out:
 - Anonymous servers : (by encryption)
Types of anonymous servers : Anonymous & pseudonymous.
 - Anonymous users:
- All anonymity types are not 100 % anonymous.
- Anybody with a basic knowledge of computing networking would know, there is always a possibility to find those who misuse the Internet.

Anonymity on the Internet

- Anonymity has its good and bad sides.
- Advantages:
 - Checking unhealthy activities within an organization.
 - National security.
 - Some relationships and security of some people.
- Disadvantages:
 - Criminals can use it to their advantage, especially in online social networks.

Anonymity Vs. Single Identity



Privacy and Data Ethics Organization

- [General Data Protection Regulation \(GDPR\)](#) European Union Regulation
- [California Consumer Privacy Act \(CCPA\)](#)